Statistical Learning and Data Analysis Module B

Cluster Analysis II: Non-Hierarchical Methods

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Statistics, tèchnè-loghìa, analysis of data

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Non-Hierarchical Methods

- Combinatorial Algorithms
- K-means Clustering
- **Representative-based Algorithms:** Mahalanlobis K-means, Kernel K-means, K-Median, K-Medoids Clustering
- *Gaussian Mixtures as Soft K-Means Clustering*
- *Soft K-means Clustering, also known as Probabilistic D-Clustering*

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Combinatorial Algorithms

- The most popular clustering algorithms directly assign each observation to a group or cluster without regard to a probability model describing the data.
- Each observation is uniquely labeled by an integer $i \in \{1, \dots, N\}$.
- A prespecified number of clusters $K < n$ is postulated, and each one is labeled by an integer $k \in \{1, \dots, K\}$.
- Each observation is assigned to *one and only one cluster*.
- These assignments can be characterized by a **many-to-one mapping** or **encoder**

  $k = C(i)$, that assigns the i-th observation to the k-th cluster.
- One seeks the particular encoder $C^*(i)$ that achieves the required goal to be defined, based on the pre-specified dissimilarities $d(x_i, x_{i'})$ between every pair of observations.

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The general idea

- The encoder $C(i)$ is explicitly delineated by giving its value (*cluster assignment*) for each observation i.
- Thus, the "parameters" of the procedure are the individual cluster assignments for each of the n observations.
- These are adjusted so as ***to minimize a "loss" (or energy) function*** *that characterizes the degree to which the clustering goal is not met.*

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The loss function

- One approach is to directly specify a mathematical loss function and attempt to minimize it through some combinatorial optimization algorithm.
- Since the goal is to assign close points to the same cluster, **a natural loss function** to be minimized would be

$$W(C) = \frac{1}{2}\sum_{k=1}^{K}\sum_{C(i)=k}\sum_{C(i')=k} d(x_i, x_{i'})$$

given the pre-specified dissimilarities $d(x_i, x_{i'})$ between every pair of observations.

- This criterion characterizes the extent to which observations assigned to the same cluster tend to be close to one another and is sometimes referred to as the *Within-Cluster Point Scatter* or *Within-Cluster Variation*.

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The Total Point Scatter Decomposition

- The *Total Point Scatter* or *Total Variation*, which is a constant given the data, independent of cluster assignment, can be defined as:

$$T = \frac{1}{2}\sum_{i=1}^{N}\sum_{i'=1}^{N} d(x_i, x_{i'}) = \frac{1}{2}\sum_{k=1}^{K}\sum_{C(i)=k}\left(\sum_{C(i')=k} d(x_i, x_{i'}) + \sum_{C(i')\neq k} d(x_i, x_{i'})\right) = W(C) + B(C)$$

where

- $W(C) = \frac{1}{2}\sum_{k=1}^{K}\sum_{C(i)=k}\sum_{C(i')=k} d(x_i, x_{i'})$ is the *Within-Cluster Point Scatter* or *Within-Cluster Variation;*
- $B(C) = \frac{1}{2}\sum_{k=1}^{K}\sum_{C(i)=k}\sum_{C(i')\neq k} d(x_i, x_{i'})$ is the *Between-Cluster Point Scatter or Between-Cluster Variation.*
- Since $W(C) = T - B(C)$, ***minimizing $W(C)$ is equivalent to maximizing $B(C)$.***

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Complex Optimization problem

- Cluster analysis by combinatorial optimization is straightforward in principle. One simply minimizes W or equivalently maximizes B over all possible assignments of the N data points to K clusters.
- Unfortunately, such optimization by complete enumeration is feasible only for very small data sets.
- The number of distinct assignments is (Jain and Dubes, 1988)

$$S(N,K) = \frac{1}{K!}\sum_{k=1}^{K}(-1)^{K-k}\binom{K}{k}k^N$$

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Sub-Optimal Partition

- For example, $S(10,4) = 34{,}105$ which is quite feasible.
- $S(N,K)$ grows very rapidly with increasing values of its arguments. Already $S(19{,}4) \simeq 1010$, and most clustering problems involve much larger data sets than $N = 19$.
- For this reason, practical clustering algorithms are able to examine only a very small fraction of all possible encoders $k = C(i)$.
- The goal is to identify a small subset that is likely to contain the optimal one, or at least *a good suboptimal partition*.

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Iterative greedy descent

- Such feasible strategies are based on ***iterative greedy descent.***
- An initial partition is specified.
- At each iterative step, the cluster assignments are changed in such a way that the value of the criterion is improved from its previous value.
- **Clustering algorithms of this type differ in their prescriptions for modifying the cluster assignments at each iteration.**
- When the prescription is unable to provide an improvement, the algorithm terminates with the current assignments as its solution.
- Since the assignment of observations to clusters at any iteration is a perturbation of that for the previous iteration, only a very small fraction of all possible assignments are examined.
- However, ***these algorithms converge to local optima which may be highly suboptimal when compared to the global optimum.***

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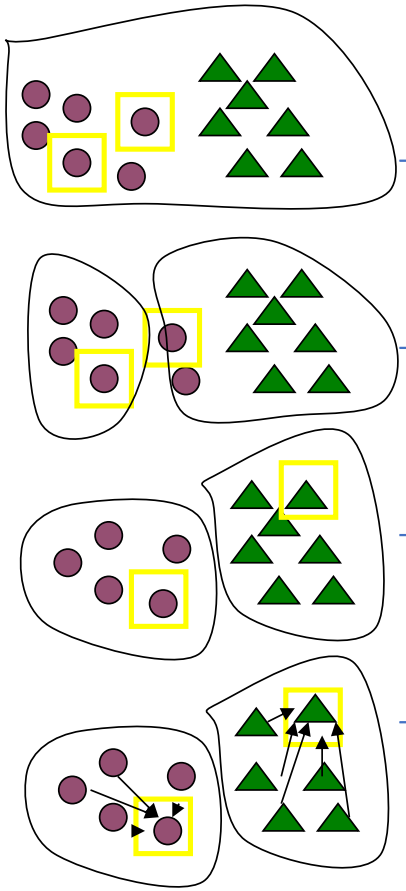
Moving Centres Method – Forgy 1965



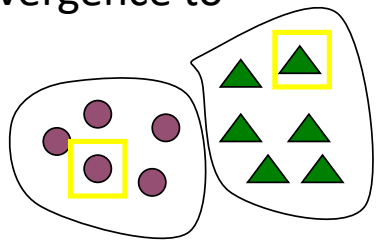
Definition of the number of classes and selection of representative units (centres of each class).

Calculation of the distances between the centre and the units attributed to the nearest seed point.

Definition of the new centres for each class considering the units closest to the centroid of the group.

Calculation of the distances between the new centres and the statistical units allocated to the nearest groups.

Repeat steps 2: 4 until to convergence to one stable solution.



K-Means Clustering

- Assume **all variables are of the quantitative type.**
- The **squared Euclidean distance** is chosen as the dissimilarity measure:

$$d(x_i, x_{i'}) = \sum_{j=1}^{d} \left(x_{ij} - x_{i'j}\right)^2 = \|x_i - x_{i'}\|^2$$

- Note that weighted Euclidean distance can be used by redefining the data values.
- The **Within-Cluster Variation** can be written as

$$W(C) = \frac{1}{2} \sum_{k=1}^{K} \sum_{C(i)=k} \sum_{C(i')=k} \|x_i - x_{i'}\|^2 = \sum_{k=1}^{K} N_k \sum_{C(i)=k} \|x_i - \bar{x}_k\|^2$$

where $\bar{x}_k = (\bar{x}_{1k}, \dots, \bar{x}_{dk})$ is the mean vector associated with the k-th cluster and $N_k = \sum_{i=1}^{N} I(C(i) = k)$.

- This criterion is minimized by assigning the N observations to the K clusters in such a way that within each cluster the average dissimilarity of the observations from the cluster mean is minimized.

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The iterative K-Means descent algorithm

- An iterative descent algorithm is used to solve:

$$C^* = \min_C \sum_{k=1}^{K} N_k \sum_{C(i)=k} \|x_i - \bar{x}_k\|^2$$

- We can obtain C^* by solving the enlarged optimization problem

$$\min_{C,\{m_k\}_1^K} \sum_{k=1}^{K} N_k \sum_{C(i)=k} \|x_i - m_k\|^2$$

named Total-Cluster Variation TCV.

- This can be minimized by an **alternating optimization procedure**. Indeed, each of the step one and two reduces TCV so that the convergence is assured.

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K-Means Clustering Algorithm

Randomly assign each observation to one of the K clusters.

Until the clustering assignment stop changing:

1. The TCV is minimized with respect to $\{m_1, \dots, m_K\}$. For each of the K clusters, compute the *cluster centroid*, or *barycenter*, or *cluster center*:

$$\min_{\{m_k\}_1^K} \sum_{k=1}^{K} N_k \sum_{C(i)=k} \|x_i - m_k\|^2$$

2. Given a current set of means $\{m_1, \dots, m_K\}$ the TCT is minimized by assigning each observation to the closest (current) cluster mean:

$$C(i) = \underset{1 \le k \le K}{\operatorname{argmin}} \|x_i - m_k\|^2$$

Stuart Lloyd and Bell Labs (1957)
Mac Queen method (1967)

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Convergence

- Each of steps 1 and 2 reduces the value of the criterion *TCV*, so that convergence is assured.
- The algorithm of Hartigan and Wong (1979) goes further and ensures that there is no single switch of an observation from one group to another group that will decrease the objective.
- When the algorithm finds a solution, there is no guarantee that the achieved solution is optimum.
- K-means algorithm can return a local optimum. For this reason, one should start the algorithm with *many different random choices* for the starting means and choose the solution having smallest value of the objective function.

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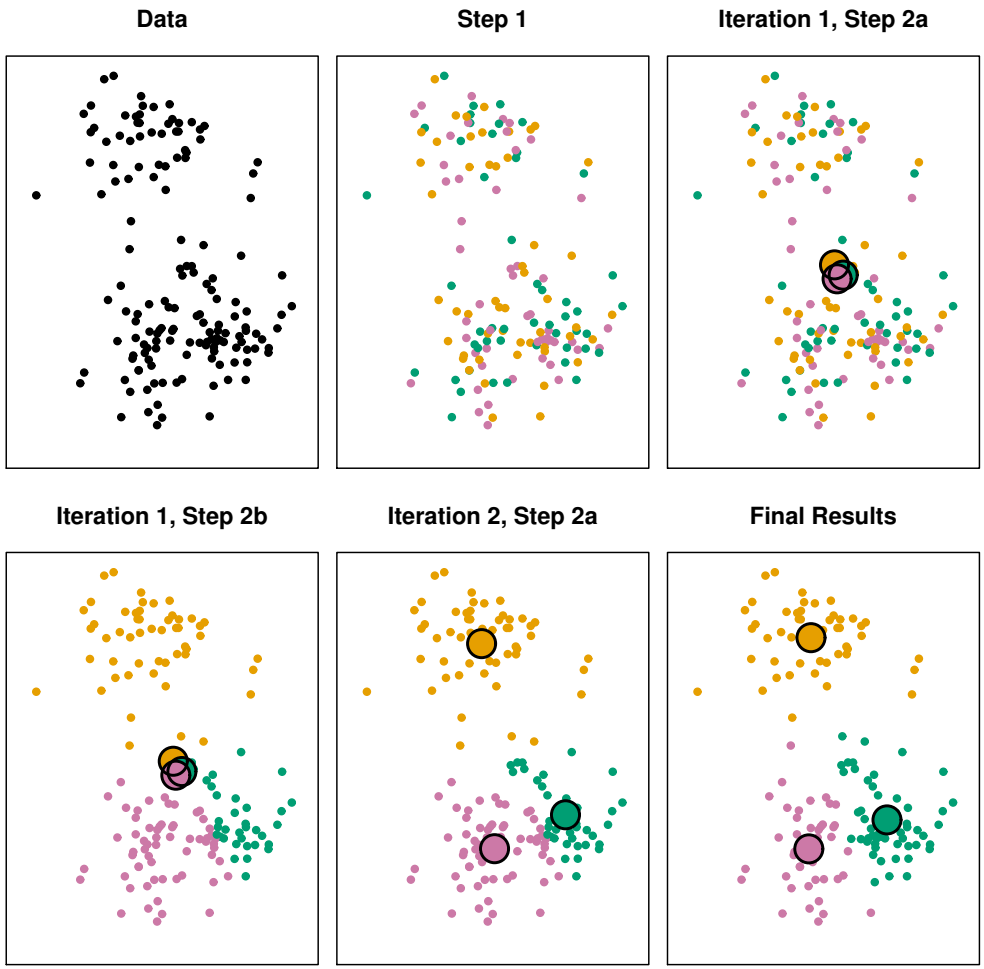
K-Means Algorithm with K=3

- The centroids are depicted by “O”s.
- A straight line can be designed to show the partitioning of points, each sector being the set of points closest to each centroid: this partitioning is called the Voronoi tessellation.



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K-Means Algorithm with K=3, different assignments

- Three different local optima were obtained, one of which resulted in a smaller value of the objective and provides better separation between the clusters.
- Those labeled in red all achieve the same best solution, with an objective value of 235.8.

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The influence of each variable

- Setting the weights to the same value for each variable does not necessarily give all features equal influence.
- The influence of the s-th variable on object dissimilarity depends upon its relative contribution to the average object dissimilarity over all pairs of observations in the data set.

$$\bar{D} = \frac{1}{N^2} \sum_{i=1}^{N} \sum_{i'=1}^{N} d(x_i, x_{i'}) = \sum_{s=1}^{p} w_s \cdot \bar{d}_s$$

- where $\bar{d}_s = \frac{1}{N^2} \sum_{i=1}^{N} \sum_{i'=1}^{N} d_s(x_{is}, x_{i's})$ is the average dissimilarity on the s-th variable.
- Setting $w_s = 1/\bar{d}_s$ would give all features equal influence in characterizing overall dissimilarity between objects.
- For the weighted squared Euclidean distance, it holds $\bar{d}_s = 2\hat{\sigma}_s^2$ being $\hat{\sigma}_s^2$ the sample estimate of the variance of the s-th variable.

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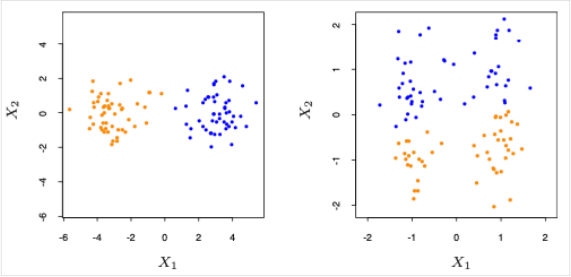
Warning

- Although this may seem reasonable, and is often recommended, it can be highly counterproductive. If the goal is to segment the data into groups of similar objects, all variables may not contribute equally to the (problem-dependent) notion of dissimilarity between objects. Some differences may reflect greater actual object dissimilarity in the context of the problem domain.
- If the goal is to discover natural groupings in the data, some features may exhibit more of a grouping tendency than others. Variables that are more relevant in separating the groups should be assigned a higher influence in defining object dissimilarity. Giving all attributes equal influence in this case will tend to obscure the groups to the point where a clustering algorithm cannot uncover them.



Standardization obscures the two well-separated groups.

Left: Raw data

Right: Standardized features, that is equivalent to use $w_s = 1/2\hat{\sigma}_s^2$

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K-Means Clustering: *Pro and Con*

- ✓ The optimizing function (Within-Cluster Variation) is always monotonic decreasing
- ✓ The number of iterations required to converge to an optimum is usually small.

But

- Finding an optimal solution is NP-hard.
- Tend to form ***convex clusters***, thus ***it cannot detect arbitrary cluster shapes.***
- ***Nearby points can end in distinct classes.***
- Assumes the squared Euclidean distance (Huygens theorem).
- It is not correct calling "k-means" algorithms that do not use Euclidean distance. It is better call them *"k-means like" algorithms*.

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Representative-Based Algorithms

- K-Means Clustering Algorithm can be understood as special case of a more general class of Clustering Algorithms called **Representative-Based Algorithms.**
- Two principles:
  - *Clusters are created on one shot;*
  - *Hierarchical relationships do not exist among different clusters.*
- This is done by using a set of ***partitioning representatives:***
  - *A function of the data points in the clusters (i.e., the mean);*
  - *Existing data points in the cluster.*

Algorithm *GenericRepresentative*(Database: $\mathcal{D}$, Number of Representatives: k)
begin
  Initialize representative set S;
  repeat
    Create clusters $(\mathcal{C}_1 \ldots \mathcal{C}_k)$ by assigning each
      point in $\mathcal{D}$ to closest representative in S
      using the distance function $Dist(\cdot,\cdot)$;
    Recreate set S by determining one representative $\overline{Y_j}$ for
      each $\mathcal{C}_j$ that minimizes $\sum_{\overline{X_i} \in \mathcal{C}_j} Dist(\overline{X_i}, \overline{Y_j})$;
  until convergence;
  return $(\mathcal{C}_1 \ldots \mathcal{C}_k)$;

- **Goal:** ***Discovering a set of high-quality clusters is equivalent to discovering a high-quality set of representatives.***
- Once the representatives have been determined, a *distance function* can be used to assign the data points to their closest representatives.

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Main Representative-Based Algorithms

- K-Means Algorithm
- *Mahalanobis K-Means Algorithm*
- Kernel K-Means Algorithm
- K-Medians Algorithm
- K-Medoids Algorithm

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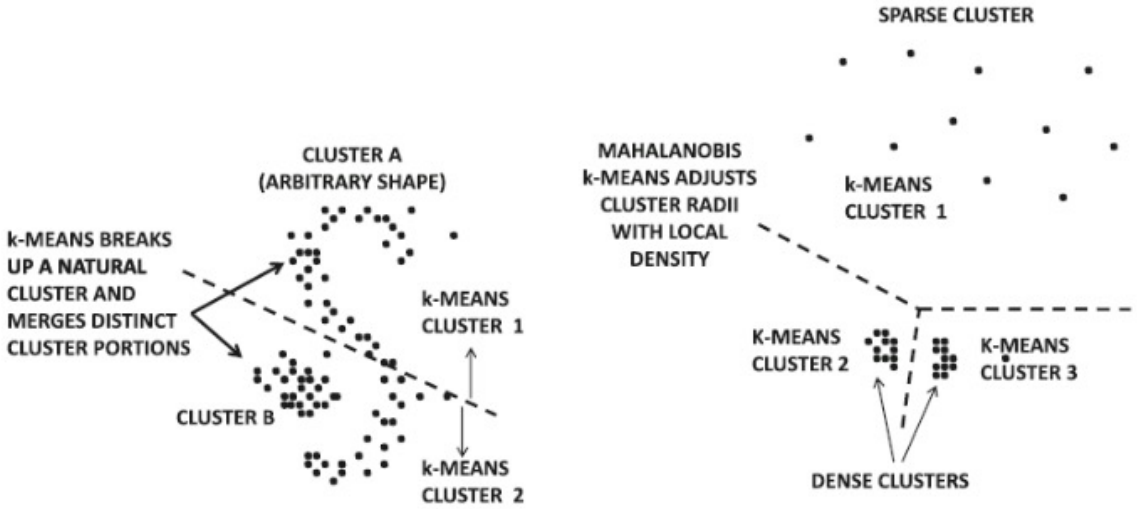
Strenghts and Weaknesses



(a) Varying cluster shape
(Bad for k-means)

(b) Varying cluster density
(Good for Mahalanobis k-means)

- k -Means is biased toward spherical clusters
- Mahalanobis k-Means can adjust well to varying cluster density

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K-Medoids Clustering Algorithm

Randomly assign each observation to one of the K clusters.

Until the clustering assignment stop changing:

1. For a given cluster assignment C, find the observation in the cluster minimizing total distance to other points in that cluster:

$$i_k^* = \underset{\{i:C(i)=k\}}{\operatorname{argmin}} \sum_{C(i')=k} D(x_i, x_{i'})$$

Then $m_k = x_{i_k^*}, k = 1, \dots, K$ are the current estimates of the cluster centers.

2. Given a current set of means $\{m_1, \dots, m_K\}$, minimize the total error by assigning each observation to the closest (current) cluster center:

$$C(i) = \underset{1 \le k \le K}{\operatorname{argmin}} D(x_i, m_k)$$

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K-Medoid Algorithm

- K-Medoids clustering is a partitioning method commonly used in domains that require
  - *robustness to outlier data,*
  - *arbitrary distance metrics,*
  - *ones for which the mean or median does not have a clear definition (such as for categorical data),*
  - *when the distance among data points is really hard to compute and computationally time consuming (i.e., Dinamic Warping Distance for time series).*
- *The center of the subset is a member of the subset, called **medoid**.*
- The K-Medoids algorithm returns medoids which are the actual data points in the data set.

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K-Medoids *versus* K-Means Clustering

✓ Representatives are selected from $\mathcal{D}$

- The representative of a K-means cluster may be distorted by outliers.
  - It can be located in an empty region which is unrepresentative of most of the data points in that cluster. Such representatives may result in partial merging of different clusters, which is undesirable.
- K-means can not be applied to heterogeneous data.
  - ***Example:*** *If the K-representatives clustering algorithm were to be applied on a set of time series of varying lengh, then how should the central representatives be defined as a function of these heterogeneous time-series? Select representatives from the original data may be very useful.*
  - Good for summarization.
- CLARANS algorithm is a scalable version of the K-Medoids for high dimensionality data.

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K-Medoids optimization

- k -Medoids is far more computationally intensive than k -means
- Optimization based on Ill-climbing strategy
  - The representative set $\mathcal{S}$ is initialized to a set of points from $\mathcal{D}$
  - $\mathcal{S}$ is iteratively improved by exchanging a single point from $\mathcal{S}$ with a point from $\mathcal{D}$
- How to perform the exchange?
  - Try all $|\mathcal{S}| \cdot |\mathcal{D}|$ possible exchanges
  - Try a randomly select set of r pairs $(\bar{X}_i, \bar{Y}_j)$ and select the best one

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Practical and Implementation Issues

- The initialization criteria
  - Select points randomly from the data space or from the data set $\mathcal{D}$
  - Sample more data points from $\mathcal{D}$ and use a hierarchical clustering approach to create k centroids.
- The choice of K
  - In practice, it is better to use large K first and then post-process
- The presence of outlier
  - Discard centers with small clusters

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Cluster Validation

Goal: to evaluate the quality of a clustering analysis

- **Internal Validation Criteria**
  - Do not need additional information
  - Biased toward one algorithm or the other
- **External Validation Criteria**
  - Ground-truth clusters are known (when using a synthetic test set or when labels are available, thus an unsupervised approach is used to a supervised approach)
  - Ground-truth may not reflect the natural clusters in the data
  - To validate the consensus of two partitions obtained by two different algorithms

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How many clusters?

- The number K of clusters must be known a-priori.
- If we do not know it, we can proceed with some tools, such as
  - ✓ Silhouette
  - ✓ GAP
  - ✓ Scree-plot
  - ✓ Calinski-Harabasz index
  - ✓ Davies-Boudin criterion

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Silhouette Coefficient

- The silhouette value for each point is a measure of how similar that point is to points in its own cluster, when compared to points in other clusters. The silhouette value for the i-th point, S_i, is defined as

$$S_i = \frac{b_i - a_i}{max\{b_i, a_i\}} \in [-1,1]$$

- where a_i is the average distance from the i-th point to the other points in the same cluster as i, and b_i is the minimum average distance from the i-th point to points in a different cluster, minimized over clusters.
- The overall Silhouette coefficient is the average $S = n^{-1} \sum_i S_i \in [-1,1]$.
- The silhouette value ranges from -1 to $+1$.
- A high silhouette value indicates that the i-th point is well-matched to its own cluster, and poorly-matched to neighboring clusters.
  - ✓ If most points have a high silhouette value, then the clustering solution is appropriate.
  - ○ If many points have a low or negative silhouette value, then the clustering solution may have either too many or too few clusters.

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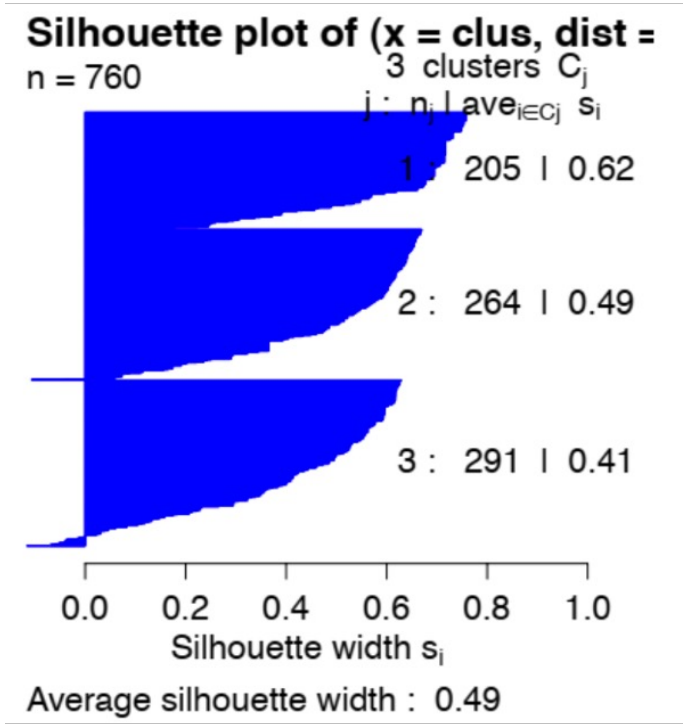
The optimal K based on Silhouette Coefficient

- A way to choose the «optimal» K is to perform the K-Means with different levels of K, then compute the clusters silhouette for each solution and choose the clustering solution with the largest averaged silhouette.
- The silhouette clustering evaluation criterion can be used with any distance metric.



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Rand Index

$\mathcal{P}$ \ $\mathcal{P}^*$	1	0	Total
1	c_{11}	c_{10}	c_{1+}
0	c_{01}	c_{00}	c_{0+}
Total	c_{+1}	c_{+0}	$\binom{n}{2}$

- Similarity between two partitions can be evaluated comparing the cluster assignment of any couple of objects in the two partitions.
- The total number of comparisons is equal to $\binom{n}{2} = \frac{n(n-1)}{2}$.
- For any partition $\mathcal{P}$ and any couple of objects we can define a Boolean variable to be equal to 1 if the two objects of the couple belongs to the same cluster and 0 if the two objects of the couple belong to distinct clusters.
- A confusion matrix allows to describe:
  - c_{11}: the number of couples of objects that belong to the same cluster in both the partitions $\mathcal{P}$ and $\mathcal{P}^*$
  - c_{10}: the number of couples of objects that belong in the same cluster in the partition $\mathcal{P}$ but in distinct clusters in the partition $\mathcal{P}^*$
  - c_{01}: the number of couples of objects that belong in distinct clusters in the partition $\mathcal{P}$ but in the same cluster in the partition $\mathcal{P}^*$
  - c_{00}: the number of couples of objects that belong in distinct clusters in both the partitions $\mathcal{P}$ and $\mathcal{P}^*$
- The RAND Index: $R_{P,P*} = \frac{c_{11}+c_{00}}{\binom{n}{2}} \in [0,1]$ is a similarity measure between the partitions $\mathcal{P}$ and $\mathcal{P}^*$. The complement to one is a dissimilarity measure.

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Main issues of Non-Hierarchical Clustering Analysis

- It can reallocate an individual that was misplaced in its cluster
- It is generally computationally efficient (i.e., K-means)
- It can improve the objective function obtained with some hierarchical methods (i.e., K-means and Ward).
- The number of clusters (or some other parameters, for model-based clustering) has to be known (or estimated) a priori.
- No taxonomic type of relationship between clusters is obtained and no dendrogram is produced.
- Some methods work only with “geometric” data and may require the Euclidean distance.

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